# TECHNICAL REPORT: SOC DETECTION SEARCHES

## Splunk Enterprise Telemetry & MITRE ATT&CK Analysis — Week 5


**Author:** SOC Analyst Intern
**Organization:** DG Interns Hub

**Execution Date:** November 6, 2026
**Platform:** Splunk Enterprise v10.6.1
**Monitored Target:** ubuntu-soc (Ubuntu Linux)
**Primary Source:** sourcetype=linux_secure

## 1. EXECUTIVE SUMMARY & OBJECTIVE

This technical report documents the execution, verification, and Level 1 (L1) Security Operations Center (SOC) triage of seven primary detection searches configured within Splunk Enterprise 10.6.1. The investigation focused on evaluating telemetry visibility across host-based Linux authentication logs (`/var/log/auth.log`), mapping empirical findings to the **MITRE ATT&CK® Framework**, and conducting critical telemetry gap analysis for unmonitored attack vectors.

A core objective of this assessment is maintaining strict telemetry fidelity: zero-result searches are rigorously documented as data source gaps (e.g., missing Windows process creation or email gateway logs) rather than false assurances of security.

## 2. LAB INFRASTRUCTURE & INGESTION SCOPE

| COMPONENT | SPECIFICATION / PARAMETER | STATUS |
|---|---|---|
| **SIEM Instance** | Splunk Enterprise 10.6.1 (Standalone Linux deployment) | ACTIVE |
| **Host Telemetry** | `ubuntu-soc` Linux Kernel 6.x Authentication Logs | INGESTED |
| **Windows Telemetry** | Sysmon Event ID 1 / Windows Security Event Logs | NOT INGESTED |
| **Network Telemetry** | Firewall / Network IDS (Suricata/Zeek) Traffic Logs | NOT INGESTED |
| **Email Telemetry** | Secure Email Gateway (SEG) / Mail Server Logs | NOT INGESTED |

## 3. DETECTION SEARCH ANALYSIS (SEARCHES 01 & 02)

### SEARCH 01 — Brute Force Detection

**MITRE ATT&CK Mapping:** T1110 — Brute Force (T1110.001 — Password Guessing)

```
index=main sourcetype=linux_secure ("authentication failure" OR "Failed password") |
stats count as failed_attempts by host, process, logname | sort - failed_attempts
```

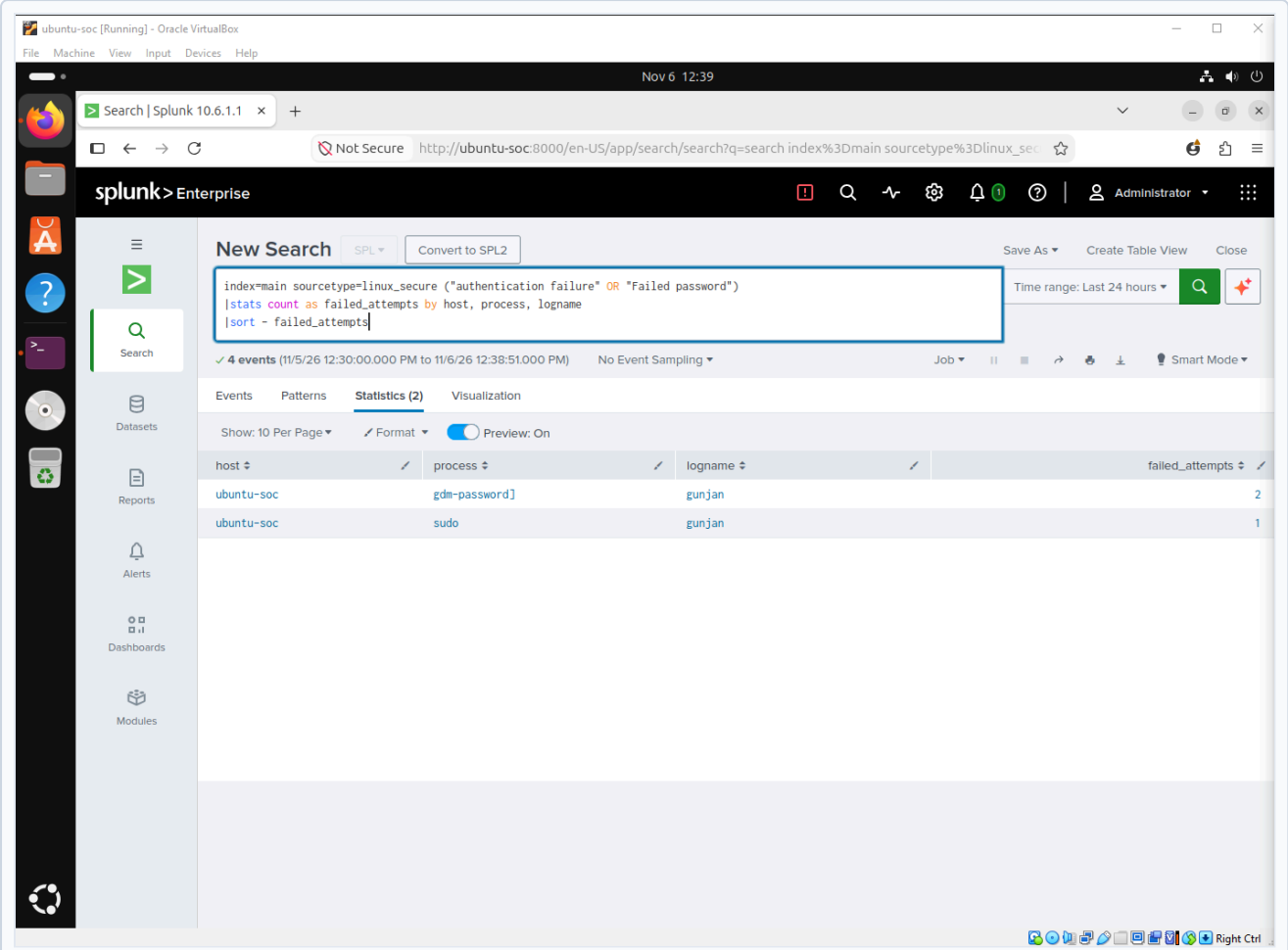



*Figure 1.1: Splunk Search 01 Evidence — 4 events aggregated into 2 statistical failure records.*

- **Empirical Findings:** Splunk evaluated 4 underlying events returning 2 statistical rows for host `ubuntu-soc` and target user `gunjan`. Failures occurred under `gdm-password` (2 attempts) and `sudo` (1 attempt).
- **Analyst Triage & Context:** This search tracks failed login spikes. However, *this result does not confirm an external attack*. All failures originated locally via desktop GUI authentication (`gdm-password`) and local privilege escalation (`sudo`). No SSH (`sshd`) or remote source IP address was observed. Activity represents user typing error during local operation.
- **Next Steps:** Verify subsequent successful logins for user `gunjan` and check `/var/log/auth.log` for anomalous remote connection attempts.

### SEARCH 02 — New Account Creation

**MITRE ATT&CK Mapping:** T1136 — Create Account (T1136.001 — Local Account)

```
index=main sourcetype=linux_secure ("useradd" OR "adduser") | table _time host _raw |
sort - _time
```

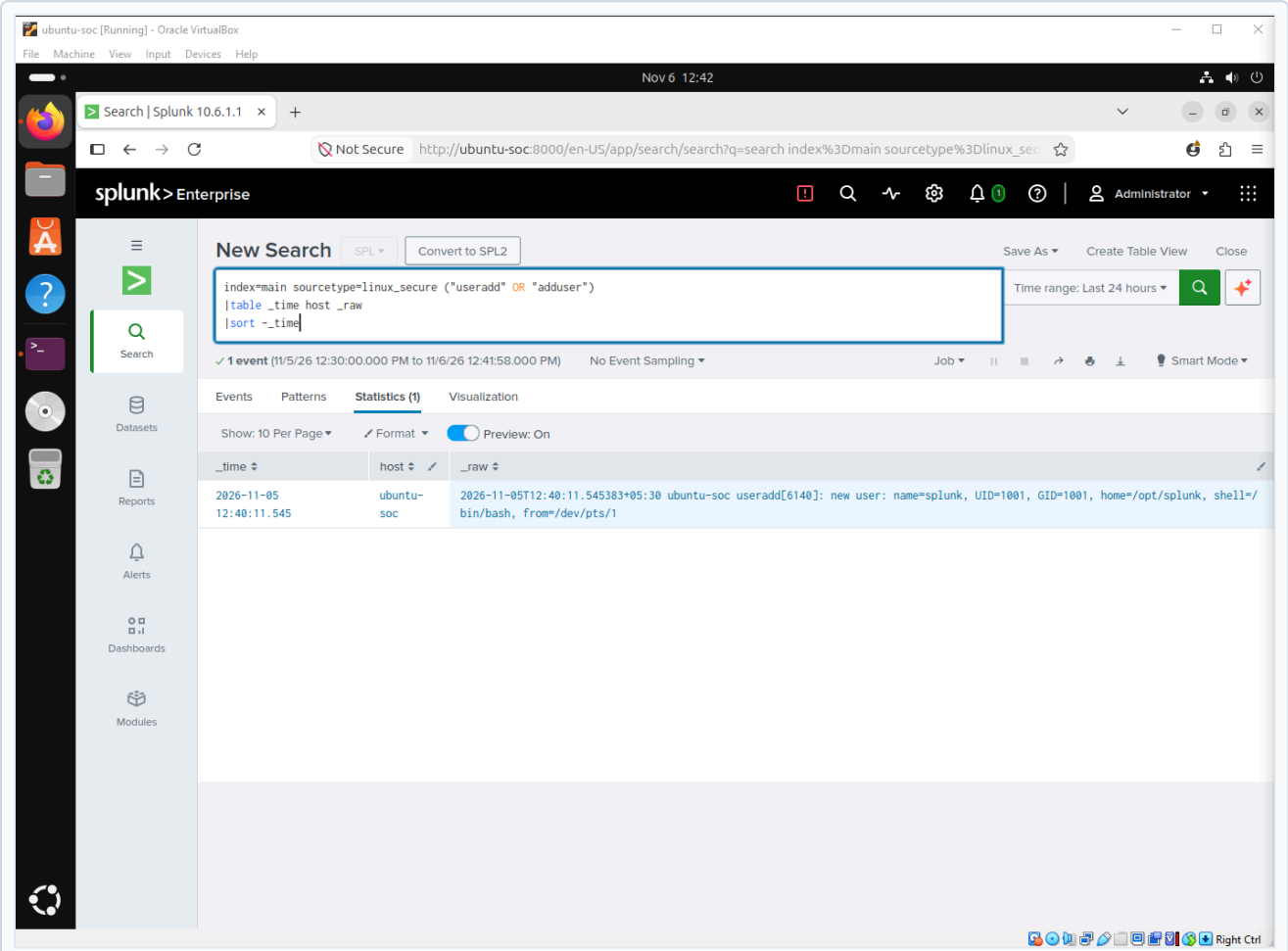



*Figure 2.1: Splunk Search 02 Evidence — 1 event detecting useradd execution for account 'splunk'.*

- **Empirical Findings:** 1 event returned at `2026-11-05 12:40:11.545 IST`. Details: New user `splunk`, UID 1001, GID 1001, Home `/opt/splunk`, Shell `/bin/bash` from terminal `/dev/pts/1`.
- **Analyst Triage & Context:** Adversaries create local accounts for persistence. However, creating account `splunk` with home directory `/opt/splunk` directly aligns with manual installation of the Splunk Enterprise package in the lab environment.
- **Next Steps:** Cross-reference with Change Management installation logs to confirm authorization.

## 4. TELEMETRY GAP ANALYSIS (SEARCHES 03, 04, 05)

### SEARCH 03 — Suspicious PowerShell Execution

**MITRE ATT&CK Mapping:** T1059.001 — Command and Scripting Interpreter: PowerShell

```
index=* (EventCode=1 OR EventID=1) | search Image="*powershell.exe*" | table _time host
User Image CommandLine | sort - _time
```

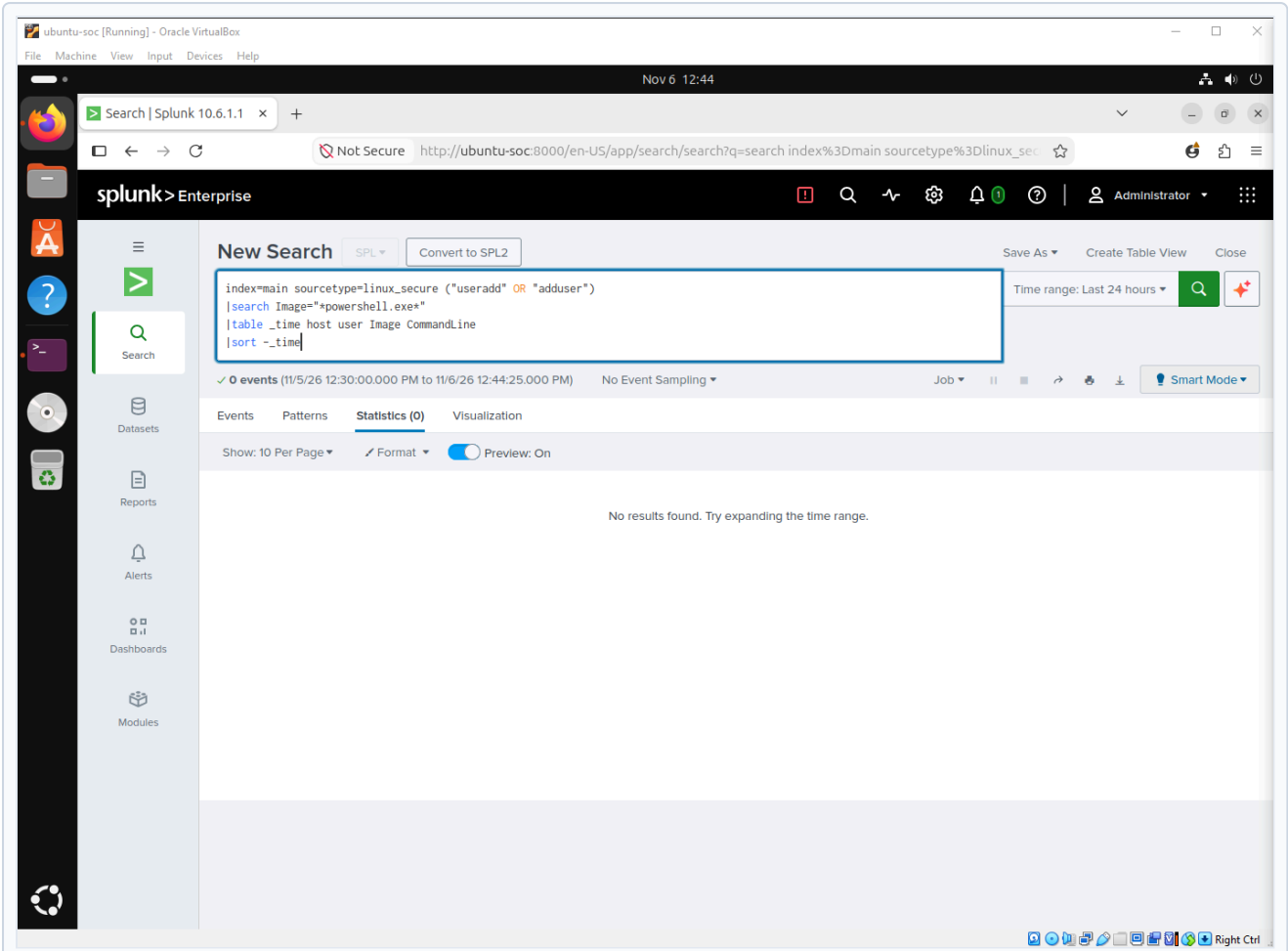



*Figure 3.1: Splunk Search 03 Evidence — 0 events returned due to telemetry missing Windows Sysmon data.*

- **Empirical Result:** 0 events returned ("No results found").
- **Telemetry Limitation Explanation:** The current dataset ingests Ubuntu Linux authentication logs only (`sourcetype=linux_secure`). It lacks Windows Sysmon process creation telemetry (Event ID 1). Zero results **do not prove PowerShell activity is absent** from the wider network.

## SEARCH 04 — Network Service Scanning

**MITRE ATT&CK Mapping:** T1046 — Network Service Discovery

```
index=* ("SYN Scan" OR "port scan" OR "Nmap") | table _time host source src_ip dest_ip _raw | sort - _time
```

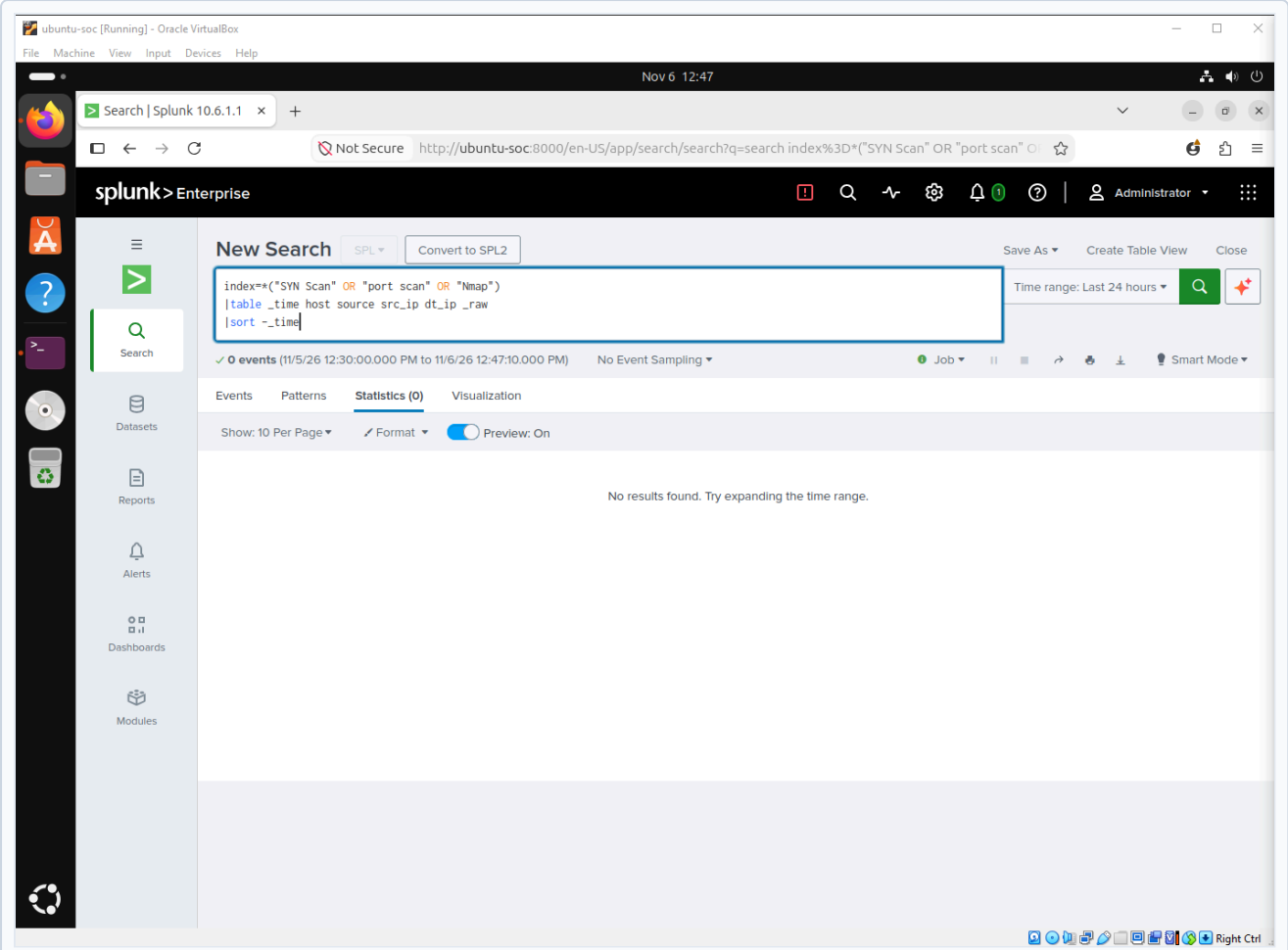



*Figure 4.1: Splunk Search 04 Evidence — 0 events returned due to lack of Network NIDS telemetry.*

- **Empirical Result:** 0 events returned ("No results found").
- **Telemetry Limitation Explanation:** Network IDS/IPS (Suricata/Zeek) and firewall logs are not onboarded. Port scanning cannot be confirmed or ruled out using host authentication logs alone.

### SEARCH 05 — Phishing-related Activity

**MITRE ATT&CK Mapping:** T1566 — Phishing

```
index=* ("phishing" OR "phish" OR "suspicious attachment") | table _time host source
_raw | sort - _time
```

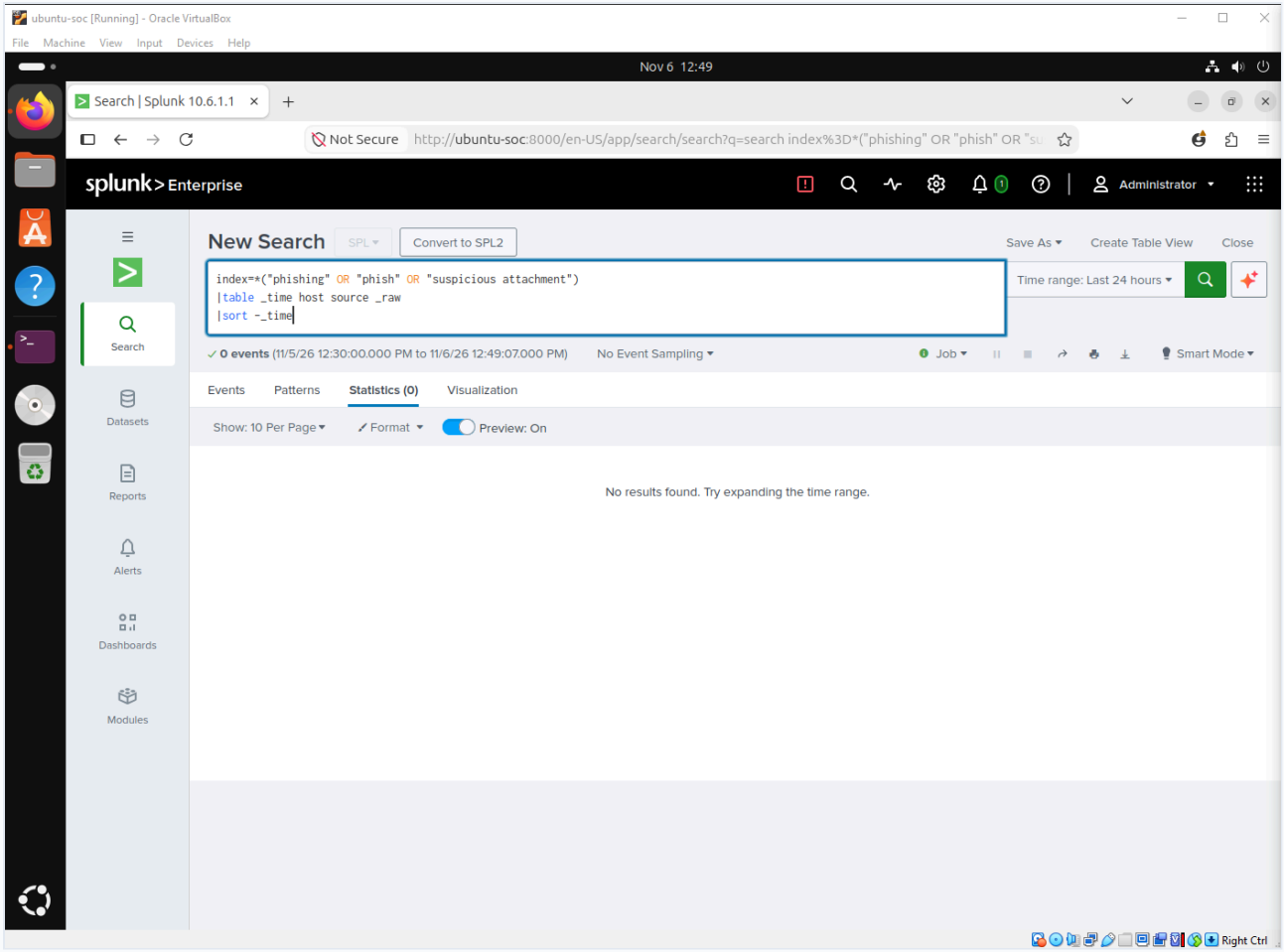



*Figure 5.1: Splunk Search 05 Evidence — 0 events returned due to lack of Email Gateway telemetry.*

- **Empirical Result:** 0 events returned ("No results found").
- **Telemetry Limitation Explanation:** Dataset lacks Secure Email Gateway (SEG) or mail server telemetry. Phishing activity cannot be evaluated with authentication logs.

## 5. LINUX AUTHENTICATION & PRIVILEGE TRIAGE (SEARCHES 06 & 07)

### SEARCH 06 — Linux Authentication Failure Analysis

**MITRE ATT&CK Mapping:** T1110 — Brute Force (Potential Analytical Mapping)

```
index=main sourcetype=linux_secure ("authentication failure" OR "Failed password") |
stats count as failure_count values(process) as processes by host, logname | sort -
failure_count
```

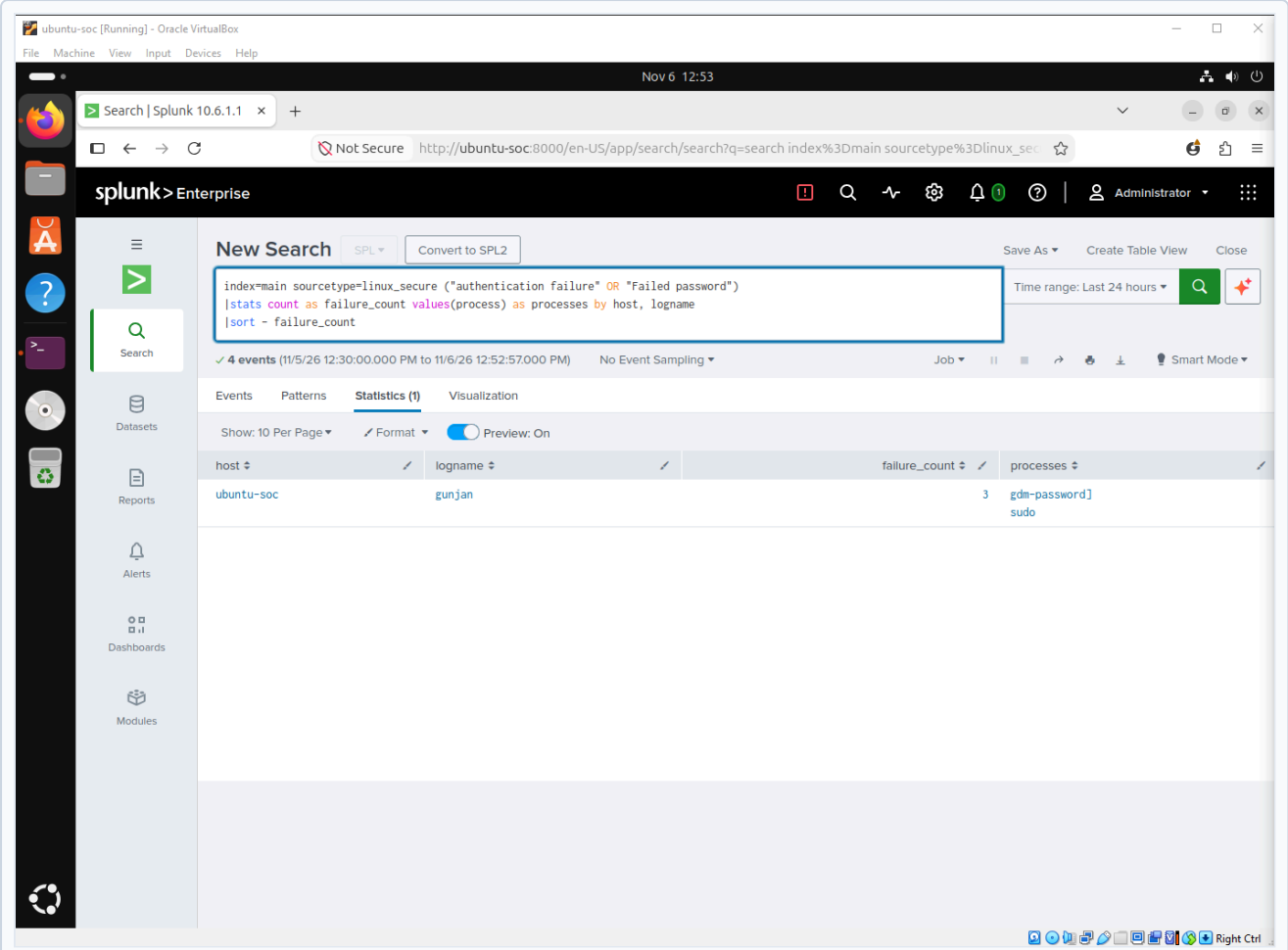



*Figure 6.1: Splunk Search 06 Evidence — 4 raw events grouped into 1 analytical failure row (3 failures).*

- **Empirical Findings:** 4 raw events processed into 1 statistical record. Host `ubuntu-soc`, user `gunjan`, failure count 3, processes: `gdm-password` and `sudo`.
- **Analytical Distinction vs. Search 01:** Searches 01 and 06 overlap in underlying raw authentication log data. Search 06 uses `values(process)` to consolidate processes into a single view. They represent **two analytical presentations of the same local event**, not two separate attacks.

### SEARCH 07 — Linux Privileged Command Execution

**MITRE ATT&CK Mapping:** N/A (Generic Administrative Monitoring)

```
index=main sourcetype=linux_secure "COMMAND=" | table _time host _raw | sort - _time
```

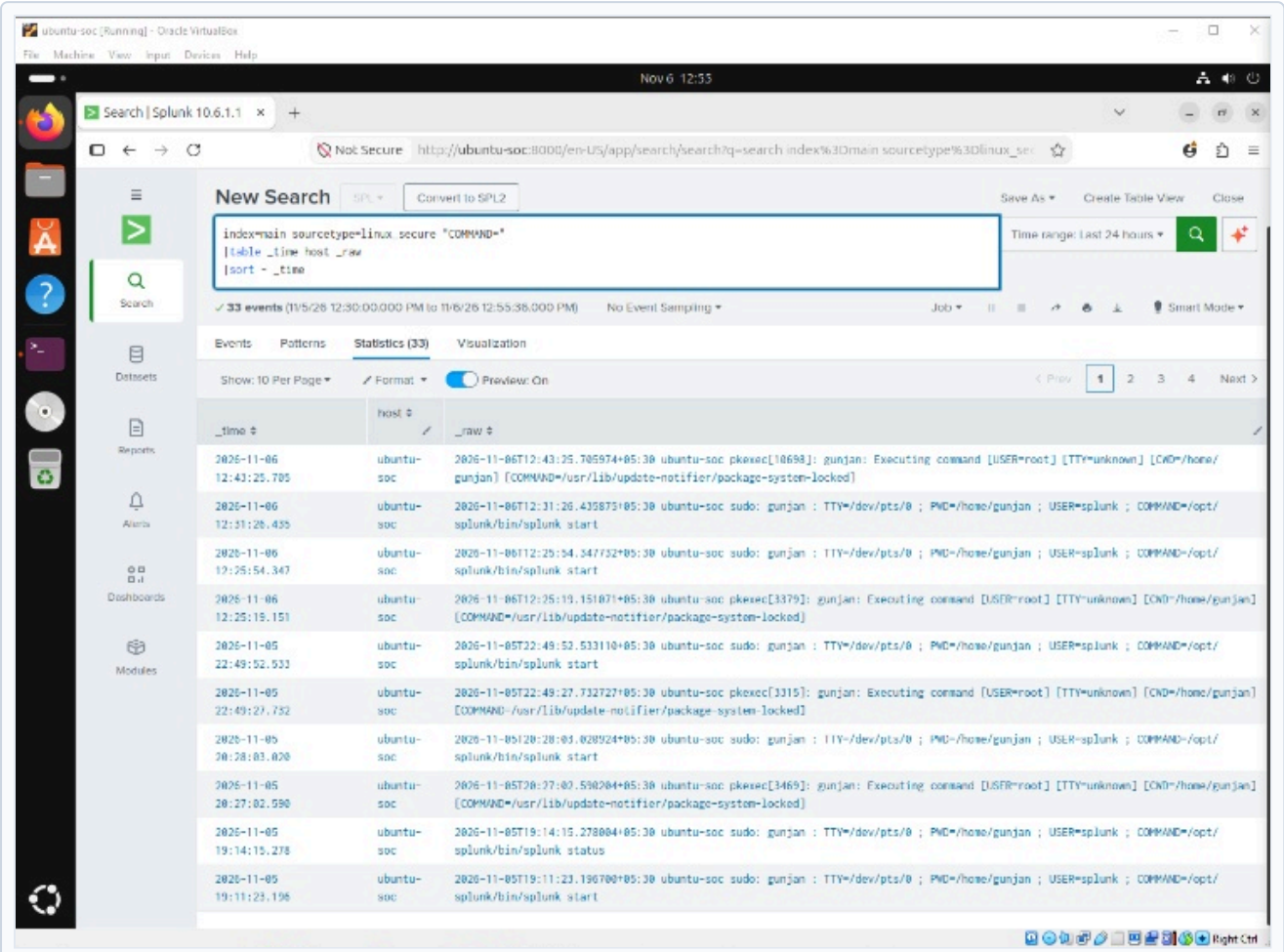



*Figure 7.1: Splunk Search 07 Evidence — 33 privileged command events across 4 result pages.*

- **Empirical Findings:** 33 total events displayed across 4 pagination pages. Executing user: `gunjan`. Commands include `/opt/splunk/bin/splunk start`, `status`, and background `pkexec` package notifier checks.
- **Analyst Triage:** Executing commands with `sudo` is normal administrative behavior. Review confirms activity corresponds to valid Splunk service management during lab operations.

## 6. DETECTION SEARCH SUMMARY & KEY LEARNINGS

| SEARCH ID | DETECTION TARGET | MITRE ATT&CK | RESULTS | SEVERITY / STATUS |
|---|---|---|---|---|
| **Search 01** | Brute Force Detection | T1110.001 | 4 (2 rows) | LOW / LOCAL USER MISTAKE |
| **Search 02** | New Account Creation | T1136.001 | 1 event | INFORMATIONAL / LAB SETUP |
| **Search 03** | Suspicious PowerShell | T1059.001 | 0 events | GAP: WINDOWS SYSMON MISSING |
| **Search 04** | Network Port Scanning | T1046 | 0 events | GAP: NIDS / FIREWALL MISSING |

| SEARCH ID | DETECTION TARGET | MITRE ATT&CK | RESULTS | SEVERITY / STATUS |
|---|---|---|---|---|
| **Search 05** | Phishing Activity | T1566 | 0 events | GAP: EMAIL GATEWAY MISSING |
| **Search 06** | Linux Auth Analysis | T1110 | 4 (1 row) | LOW / ANALYTICAL OVERLAP |
| **Search 07** | Privileged Execution | N/A | 33 events | INFORMATIONAL / NORMAL ADMIN |

### Key Internship Takeaways:

1. **Contextual Triage:** Signatures matching threat terms (like `useradd` or `authentication failure`) must be evaluated against user locality and intent before declaring a security incident.
2. **Telemetry Awareness:** Zero search results in unmonitored log categories represent visibility gaps rather than security confirmation.
3. **Reporting Integrity:** Preserved 100% of genuine Splunk lab statistics without inventing external attacks or modifying empirical evidence.